

NEELESH MISHRA

SENIOR SOFTWARE ENGINEER

CONTACT

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CORE STACK

React / Next.js
Node.js / TypeScript
PHP / Laravel / CodeIgniter
Tailwind / MUI / shadcn/ui
PostgreSQL / MySQL / MongoDB

CLOUD & DELIVERY

AWS (Certified)
Docker
CI/CD pipelines
Prisma ORM

APPLIED AI

Retrieval-Augmented Generation
Azure OpenAI
Vector search / pgvector
Local embeddings / FastEmbed
RBAC-aware retrieval

CERTIFICATIONS

AWS Certified
Davra IoT Platform Certification

EDUCATION

MCA
Mangalayatan University, 2024
B.Sc. Mathematics
CSJM University, 2014-2017

LANGUAGES

Hindi - Full Professional
English - Professional Working

PROFILE

Senior full stack developer with 8+ years building government and enterprise platforms using React, Next.js, Node.js, PHP, and cloud delivery practices. Hands-on with citation-grounded RAG, local embeddings, pgvector retrieval, Azure OpenAI, and authorization-aware AI systems. Currently delivering secure, multilingual products at e& enterprise in Dubai.

EXPERIENCE

Senior Software Engineer | e& enterprise

November 2022 - Present | Dubai, UAE

- ◆ **Oyoon - Dubai Police Camera Monitoring**
Built the administration platform controlling officer access to camera feeds and locations, with user and group management, granular permissions, Active Directory SSO, ESB notifications, and complete audit history.
Next.js | TypeScript | Tailwind CSS | Active Directory | ESB
- ◆ **SmartKhateeb - AWQAF**
Built trilingual scheduling, evaluation, recording, and reporting workflows supporting 1,000+ khateeb across all seven UAE Emirates.
Next.js 15 | shadcn/ui | TanStack Table | Zustand | RTL/i18n
- ◆ **Sharjah Safari Website**
Developed a bilingual custom WordPress experience with 15+ ACF Pro content blocks, WPML localization, and an interactive park map.
WordPress | PHP | ACF Pro | WPML | MySQL
- ◆ **MPOS - Abu Dhabi Commercial Bank**
Delivered dual portals for merchant requests, POS inventory, SLA visibility, and field-team assignment workflows.
Next.js 14 | MUI | React Hook Form | Redux | Framer Motion
- ◆ **ADNOC NOC Portal**
Shipped a permitting portal for smart infrastructure across 72 ADNOC sites in all seven Emirates, including a five-stage approval chain and Mapbox installation view.
Next.js | MUI | Mapbox | ApexCharts | RTL/i18n
- ◆ **NHRI Website & Complaint Portal**
Built the bilingual public website, CMS administration, and structured citizen complaint experience for the UAE National Human Rights Institution.
Laravel | PHP | Tailwind CSS | MySQL | RTL/i18n
- ◆ **Oyoon SIM & Router Management**
Delivered a four-role hardware logistics platform with two-stage approvals, live availability, overdue alerts, and issuance and return tracking.
Next.js 15 | TypeScript | Prisma | PostgreSQL | shadcn/ui | Docker
- ◆ **GCGRA Self-Exclusion Portal**
Built an accessible national self-exclusion journey with UAE Pass authentication, five gaming categories, six-month to five-year periods, and WCAG 2.1 AA-aligned interfaces.
Next.js 15 | Tailwind CSS | Radix UI | UAE Pass | WCAG

AI TOOLKIT

Document ingestion
Metadata-aware chunking
BGE / sentence-transformers
HNSW vector retrieval
Grounded generation
Page & section citations
JWT authentication
Database-enforced RBAC
Streaming responses
Role-tagged batch ingestion

ARCHITECTURE

Next.js
FastAPI
PostgreSQL / pgvector
Azure OpenAI
Docker Compose
Alembic migrations
pytest / Jest

PRINCIPLES

Citations over unsupported claims
Permissions before model context
Local embeddings where practical
Prototype scope stated clearly

AVAILABILITY

UAE, India, and international
remote roles
neeleshmishra.dev

AI INITIATIVES

Personal, publicly shareable prototypes. Project names link to the source repositories.

Enterprise RAG | Document-Grounded Q&A | 2026

- Built a full-stack RAG prototype supporting PDF, DOCX, and TXT ingestion, metadata-aware chunking, local BGE embeddings, pgvector HNSW retrieval, and Azure OpenAI answers with page and section citations.
- Containerized the Next.js, FastAPI, and PostgreSQL stack with automatic migrations and backend and frontend test suites.

Next.js | FastAPI | PostgreSQL | pgvector | FastEmbed | Azure OpenAI | Docker

Enterprise RBAC RAG | Authorization-Aware Retrieval | 2026

- Extended the RAG architecture with JWT authentication and database-enforced RBAC inside vector search, preventing unauthorized document content from entering retrieved context or citations.
- Added streamed answers, role-tagged batch ingestion, immediate account and role revocation, and user and role administration using Next.js 15, asynchronous FastAPI, PostgreSQL/pgvector, local sentence-transformers, and Azure OpenAI.

Next.js 15 | Async FastAPI | PostgreSQL | pgvector | Sentence Transformers | Azure OpenAI

EARLIER EXPERIENCE

Software Developer | Appventurez

June 2021 - November 2022 | Noida, India

- Developed full-stack web and mobile-backend solutions for client projects using React.js, Node.js, and modern JavaScript frameworks.
- Built and integrated REST APIs connecting front-end applications with backend services and third-party platforms.
- Worked directly with clients and project managers to translate business requirements into technical specifications.

PHP Developer | TechGropse Pvt. Ltd.

October 2017 - May 2021 | Noida, India

- Developed and maintained dynamic web applications using PHP frameworks including Laravel and CodeIgniter.
- Designed and optimized MySQL database schemas and queries to support application features and reporting.
- Built responsive front-end interfaces for cross-browser compatibility, growing into increasing ownership of features over more than three years.

DELIVERY PROFILE

Secure enterprise workflows | multilingual and RTL interfaces | accessible product systems | cloud and container delivery | frontend-to-backend ownership

AI ARCHITECTURE EVIDENCE

◆ Grounded retrieval path

Document parsing and page or section metadata flow into local embeddings and pgvector HNSW search; Azure OpenAI receives only retrieved context and returns answers with source citations.

Ingestion -> chunk metadata -> local vectors -> HNSW retrieval -> grounded answer -> citations

◆ Authorization boundary

JWT identity and current database role grants constrain the vector-ranking query itself, so unauthorized content cannot enter model context or citation output.

Identity -> active role lookup -> permitted documents -> ranked context -> streamed answer

◆ Prototype operations

Docker Compose, automatic migrations, test suites, content-hash refresh, and immediate account and role checks make both initiatives repeatable and inspectable.

Containers | migrations | backend and frontend tests | revocation checks